IARE
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AgriTech — FieldReady: Early Crop Disease Detection and Action Support

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Problem Statement

DOMAIN : AgriTech

Crop diseases can spread before farmers recognize their symptoms. This may result in loss of yield and thus cause loss to the farmers. These Crop Diseases are caused by most common factors like weather conditions, soil status and other external factors. The current tools lack the accessibility in low network/internet areas and do not work well with low clarity cameras. There is a need for a tool that analyses a photograph of a crop leaf, identifies visible signs of disease and helps farmers decide what to inspect next under these conditions.

Target users: Small and marginal farmers with less interaction in technical aspects

Idea/Solution

FieldTech : Early crop disease Detection and Action support

- The farmer takes and image of leaf and uploads to the application.
- The application uses CNN as well as HSV color-space segmentation to process the image.
- The image description is compared to other images in the Database and disease is evaluated.
- Later certain predefined follow up questions are asked based on the answers of farmer the other factors are weighed.
- A planned timeline based on detected disease in given to the farmer in order to prepare a plan of action.
- Certain alerts are also given incase of certain changes in weather which are related to the crop disease.

Follow-up questions to the farmer(illustrative example):

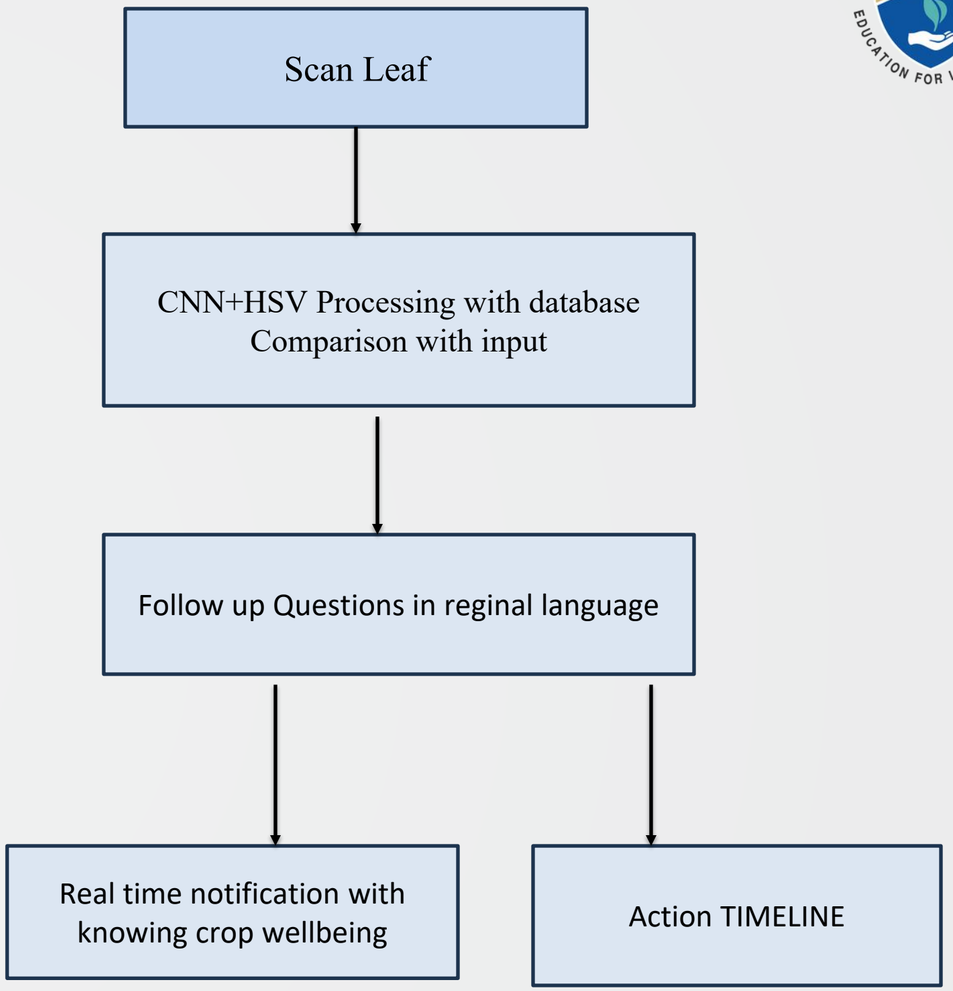
- What is the age of crop?
- How many acres is the crop spread out in?
- Any other supporting crops grown with the main one?
- Any signs of weed plants?

Spoken question and reply (Telugu / Hindi / English): the plan and timeline are read aloud

Alert example: "Heavy rain in 48 hours: delay fertilizer, drain the field."

Technical Approach

Frontend : React.js/ReactNative, Tailwind CSS, TypeScript, Next.js
Image Processing : CNN(ResNet-50,101/mobileNet)+ HSV color-space segmentation
Backend – FastAPI, TensorFlow, python libraries (numpy,pandas)
Database : SQLlite/MySQL, OpenweatherMap
Translation : Bashini (govt e-portal)
Voice Output : gTTs (google text to speech converters)



Feasibility and expected Impact

Works well in low internet areas since the application will be a PWA and has offline database resource with predefined questionnaire and CNN models

Works well with native language speaking farmers as the voice over works with multiple languages.

Also takes the user's feedback after disease detection in order to work as an adaptive model rather than a static model.

Realtime alerts based on weather conditions and real time monitoring and has real time interaction with the users.

THANK YOU